PARTIALLY INCOMPLETE. The gap-fill pass added, from search extracts only: Mata v. Avianca (June 2023 sanctions); Sculley et al. 2015; Autor 2014; the Process Mining Manifesto; an academic conformance case (ASML); vendor process-mining cases quantifying the documented-vs-actual gap; peer-reviewed RPA knowledge-loss studies; the Narayanan-Kapoor critique (via secondary coverage). Still missing: any study that causally links process-documentation quality to AI or automation success (none found, and the thesis's central mechanism remains unmeasured); peer-reviewed multi-organisation conformance rates; the exact Mata fine split; the AI Snake Oil post itself. Why: during this session the network proxy refused almost every host (CanLII, arXiv, RAND, The Register, Bloomberg, NBER, NeurIPS, METR's site, Wikipedia and others), WebFetch failed on every host tried, and the shared web-search budget ran out partway through. Most findings below rest on search-engine summaries of the cited pages, not on pages fetched in full. They are flagged as such. The only primary sources read directly were three official GitHub repositories (tau-bench, TheAgentCompany, METR), via raw.githubusercontent.com.

Track 5: What goes wrong

Last updated: 2026-11-06. Labels: EVIDENCE = measured/primary data; FORECAST = projection; OPINION = argument, vendor or consultancy view.

Verification note (added 2026-11-06): page fetches were blocked for most hosts this session, so findings marked search summary, search extract or snippet rest on search-engine extracts of the cited URL. Quotation marks inside findings mark wording as reported by that source; only quotes in this file's "Verbatim quotes" section (and in 13-quotes.md) were checked against a fetched page. Corrections from the verification pass are in verification.md.

Reading note on sourcing. For each item, "Seen via search summary" means the URL was returned by the search engine and the claim matches the engine's summary of that page, but the page itself was not opened. Treat those claims as leads to re-verify, not as checked facts. "Fetched" means the raw page was downloaded and read.

Summary

- **The named failures split into two groups, and only one fits the thesis.** Group one is organisational: unclear problem, governance, overreach. Examples: MD Anderson/Watson, Klarna, CBA, and Zillow in part. Group two is the technology itself failing: hallucinated facts, prompt injection, agents doing destructive things, poor speech accuracy. Examples: Air Canada, NYC MyCity, Deloitte, the lawyer sanctions, EchoLeak, Replit, the PocketOS deletion, McDonald's/IBM. Group two is at least as large as group one, which cuts against "the tech is the easy part."
- **The courts treat AI output as the company's own speech.** In Moffatt v. Air Canada (BC Civil Resolution Tribunal, 14 Feb 2024), the tribunal rejected Air Canada's argument that its chatbot was a separate legal entity responsible for its own actions. This is a small-claims ruling with no binding precedent, but it is the one everyone cites. Mobley v. Workday (N.D. Cal., 16 May 2025) conditionally certified a nationwide age-discrimination collective against an AI screening vendor.
- **Hallucinated legal citations are now a measured, growing stream, not an anecdote.** Damien Charlotin's database (HEC Paris) was reported at about 1,598 decisions by 9 June 2026, roughly 8 new cases a day. This figure comes from a third-party write-up and was not

checked against the live database. About half of the cases reportedly involve self-represented litigants, not lawyers.

- **Reversals after over-automating customer service are now a pattern.** Klarna (2025, CEO cited "lower quality"), Commonwealth Bank of Australia (Aug 2025, 45 roles reinstated after a Fair Work Commission challenge), McDonald's ending its IBM drive-thru test (June 2024, about 100 stores), and Taco Bell rethinking its voice AI (Aug 2025, 500+ stores). In each case the stated cause was quality or accuracy, not staff resistance.
- **Agent reliability falls fast when the same task is repeated.** On tau-bench retail, the best listed model drops from a 69% success rate on one try to 46% when it must succeed on all four tries (pass^4). GPT-4o falls from 60% to 38%. In my own refit of METR's public run data, the task length a model handles 80% of the time is 4 to 10 times shorter than its headline 50% horizon. The frontier 80% horizon is about an hour, against 12 hours at 50%.
- **The benchmarks themselves needed repair.** Sierra's τ³-bench release lists "75+ task fixes" to its own earlier tasks. When the measuring instrument is this fragile, the evaluation gap inside a client business is likely to be wider.
- **The viral "30 to 50% of RPA projects fail" figure traces to one 2016 EY thought-leadership PDF.** It says "as many as" 30 to 50% of initial RPA projects fail, and frames that as a planning failure rather than a technology failure. Later citations drop the hedge and misdate it. RAND's "more than 80% of AI projects fail" (2024) is itself attributed only to "some estimates."
- **The documented-vs-actual process gap is real, but its causal role in AI failure is unmeasured.** Process mining exists because documented processes diverge from executed ones. One vendor case found 50-60% conformance and 600+ variants of one process, and an academic ASML case found the real process much more complicated than the reference model (paraphrased). No study found links that gap to AI or automation failure rates. Autor (2014) supplies the theory (tacit knowledge) and also the counter (machine learning learns from examples, not rules).
- **RAND's interview study (65 practitioners, 2024) gives the thesis its best support.** Its first root cause is that stakeholders misunderstand or miscommunicate the problem to be solved. The remaining causes are data, infrastructure, and applying AI to problems too hard for it. Only the first is a "business not ready" cause in the thesis's sense.

Key findings

Tribunal and court rulings

- **[EVIDENCE]** Moffatt v. Air Canada, 2024 BCCRT 149, decided 14 Feb 2024 by tribunal member Christopher Rivers. Air Canada's website chatbot told a bereaved customer he could claim a bereavement fare retroactively. That contradicted the airline's actual policy page. Air Canada argued the chatbot was "a separate legal entity that is responsible for its own actions." The tribunal called this a "remarkable submission," held the chatbot was part of Air Canada's website, and found negligent misrepresentation. The customer received the fare difference plus interest and fees (small-claims jurisdiction, up to C$5,000; exact award not verified this session). CRT decisions do not bind other tribunals. Seen via search summary of: McCarthy Tétrault, Feb 2024 https://www.mccarthy.ca/en/insights/blogs/techlex/moffatt-v-air-canada-misrepresentation-ai-chatbot; Law360 Canada https://www.law360.ca/ca/articles/1804075. Primary text at CanLII (blocked this session): https://www.canlii.org/en/bc/bccrt/doc/2024/2024bccrt149/2024bccrt149.html

- **[EVIDENCE]** Mobley v. Workday (N.D. Cal.). On 16 May 2025, Judge Rita Lin granted preliminary (conditional) certification of a nationwide Age Discrimination in Employment Act collective. It covers applicants aged 40+ rejected through Workday's AI screening since September 2020. The court found applicants "similarly situated" because the claim rests on one unified policy: Workday's AI scoring, sorting, ranking or screening. Law-firm summaries report that Workday stated in filings that about 1.1 billion applications were rejected using its tools during the period. That is a scale figure, not a finding of discrimination. Workday calls the case meritless, and this is only step one of a two-step process. Developments after mid-2025 were not checked. Seen via search summary of: Holland & Knight, 27 May 2025 https://www.hklaw.com/en/insights/publications/2025/05/federal-court-allows-collective-action-lawsuit-over-alleged; Fisher Phillips https://www.fisherphillips.com/en/insights/insights/discrimination-lawsuit-over-workdays-ai-hiring-tools-can-proceed-as-class-action-6-things; HR Dive https://www.hrdive.com/news/workday-ai-bias-lawsuit-class-collective-action/748518/

- **[EVIDENCE, weak: secondary count]** The AI Hallucination Cases database, maintained by Damien Charlotin (Senior Research Fellow, HEC Paris), records decisions where a court or tribunal found that AI-hallucinated content had been submitted. It has run since April 2023. A third-party audit post reports 1,598 cases as of 9 June 2026, an implied rate of just under 8 new cases a day. Earlier counts: about 120 (May 2025), 719 (January 2026, per Social Science Space as summarised), 1,227 (early 2026), 1,545 (4 June 2026). About half reportedly involve pro se (self-represented) litigants. None of these counts were checked against the live database at damiencharlotin.com/hallucinations. Seen via search summary of: HAQQ https://haqq.ai/blog/ai-legal-hallucination-audit.html; tinyapps.org, June 2026 https://tinyapps.org/blog/ai-hallucinations-in-court.html; MSBA, 27 July 2026 https://www.msba.org/site/content/News-and-Publications/News/General-News/Confronting_the_Hard%20Numbers_Behind_Lawyers_AI_and_Phantom_Precedents.aspx; Simon Willison, 25 May 2025 https://feeds.simonwillison.net/2025/May/25/ai-hallucination-cases/

- **[EVIDENCE] Mata v. Avianca, Inc.**, No. 1:22-cv-01461 (S.D.N.Y.), Judge P. Kevin Castel, Opinion and Order on Sanctions, 22 June 2023 (gap-fill pass, search extract).

    - Lawyers Steven Schwartz and Peter LoDuca of Levidow, Levidow & Oberman filed a brief citing cases invented by ChatGPT. Avianca's lawyers told the court in March 2023 that they could not find several of the cited cases.
    - The court imposed sanctions under Rule 11, or in the alternative its inherent power. It found subjective bad faith, partly because the lawyers kept standing by the fake cases after being challenged.
    - The court fined them $5,000. Reports conflict on whether that was $5,000 in total, joint and several on the two lawyers and the firm, or $5,000 each. Most outlets report one $5,000 penalty, but check the order.
    - The lawyers also had to notify their client and each judge falsely named as the author of a fabricated opinion.
    - The opinion says that using AI for assistance is not in itself improper. The failure was not verifying, and then not correcting.
    - The underlying personal-injury claim was dismissed separately as time-barred.

Primary: https://law.justia.com/cases/federal/district-courts/new-york/nysdce/1:2022cv01461/575368/54 ; copy at https://www.courts.ri.gov:443/PDF/Mata%20v.%20Avianca%20Inc.pdf (search extracts; not fetched)

    - Relevance. This is the origin case for the hallucination-sanctions stream counted in the Charlotin database below. It is a verification failure by competent professionals using a general tool. Workflow description and shared context would not have prevented it. A check step would have.

- **[EVIDENCE]** Deloitte Australia, October 2025. The firm agreed to refund the final instalment of a A$440,000 contract with the Department of Employment and Workplace Relations (DEWR) for a welfare-compliance (Targeted Compliance Framework) review. The original report contained:

    - 12 references to a non-existent report attributed to a University of Sydney law professor
    - references to a non-existent report by a Swedish professor
    - a fabricated quote attributed to a Federal Court judge

  The corrected version discloses that a GPT-4o-based tool chain hosted on DEWR's Azure tenancy was used. DEWR and Deloitte say the findings and recommendations are unchanged. Labor Senator Deborah O'Neill criticised the partial refund. The exact refund amount was not confirmed. Seen via search summary of: The Register, 6 Oct 2025 https://www.theregister.com/2025/10/06/deloitte_ai_report_australia/; ACS Information Age https://ia.acs.org.au/article/2025/deloitte-to-refund-government-over-ai-errors.html; Cyber Daily https://www.cyberdaily.au/government/12737-deloitte-to-refund-government-after-using-ai-in-440-000-report

Customer-facing chatbots and voice AI

- **[EVIDENCE]** NYC MyCity chatbot (Microsoft Azure-based, launched under Mayor Adams). Testing by The Markup and THE CITY, first reported 29 March 2024, found it gave advice that would break the law. For example, it said landlords could refuse Section 8 vouchers (source-of-income discrimination is illegal in NYC), that employers could take workers' tips, and it gave wrong answers on tenants' rights. A 2 April 2024 follow-up found it still live. The city added disclaimers rather than taking it down. TechRadar later reported that Mayor Zohran Mamdani was "set to kill off" the bot as "functionally unusable." The date of that report and whether shutdown happened were not confirmed. Seen via search summary of: The Markup, 2 Apr 2024 https://themarkup.org/news/2024/04/02/malfunctioning-nyc-ai-chatbot-still-active-despite-widespread-evidence-its-encouraging-illegal-behavior; OECD AI incident entry https://oecd.ai/en/incidents/2024-03-29-3dce; TechRadar https://www.techradar.com/pro/zohran-mamdani-is-set-to-kill-off-new-yorks-functionally-unusable-business-chatbot-which-often-gave-out-illegal-advice
- **[EVIDENCE]** McDonald's / IBM automated order taking. The test ran in about 100 US restaurants from the October 2021 partnership. An internal franchisee email from Chief Restaurant Officer Mason Smoot said it would be switched off by 26 July 2024. McDonald's had a stated target of 95%+ order accuracy before broader rollout. Unnamed sources told CNBC the system struggled with accents and dialects. McDonald's said voice ordering would still be "part of our restaurants' future." Seen via search summary of: Nation's Restaurant News, June 2024 https://www.nrn.com/quick-service/mcdonald-s-is-ending-its-ai-drive-thru-test-with-ibm; Restaurant Business https://www.restaurantbusinessonline.com/technology/mcdonalds-ending-its-drive-thru-ai-test

- **[EVIDENCE]** Taco Bell voice AI, reported August 2025 from a Wall Street Journal interview. The system had run in more than 500 US drive-thrus since 2023 and handled over 2 million orders. Viral prank orders, including one for 18,000 water cups, and looping prompts led to a reassessment. Chief Digital and Technology Officer Dane Mathews described a restaurant-by-restaurant approach that keeps humans in the loop at peak times. Some outlets called this a "pause," but the WSJ-based reporting describes a hybrid rollout, not a halt. The WSJ article itself was not reached. Seen via search summary of: Jalopnik https://www.jalopnik.com/1956939/taco-bell-drive-through-18000-waters/; AI Incident Database https://incidentdatabase.ai/reports/6654
- **[EVIDENCE]** Klarna. In February 2024 Klarna said its AI assistant did the work of about 700 agents. Coverage gives the share of chats handled variously as two-thirds and 75%; Klarna's own release was not reached. In May 2025, CEO Sebastian Siemiatkowski told Bloomberg that Klarna would make sure customers could always reach a human. He said the AI approach had produced "lower quality" service and that cost had been too dominant a factor. Caveats from Computing: the new hires are freelance roles in an outsourcing arm, Klarna still pursues headcount reduction through attrition, and AI still handles simpler queries. Seen via search summary of: Computing, 2025 https://www.computing.co.uk/news/2025/ai/meat-back-on-the-menu-as-klarna-rows-back-on-ai; Entrepreneur https://www.entrepreneur.com/business-news/klarna-ceo-reverses-course-by-hiring-more-humans-not-ai/491396; CX Dive https://www.customerexperiencedive.com/news/klarna-pursues-uber-style-customer-service-model/812763/
- **[EVIDENCE]** Commonwealth Bank of Australia, August 2025. CBA cut 45 direct-banking roles, citing a new voice bot that it said had reduced call volumes by about 2,000 a week. The Finance Sector Union (FSU) took the matter to the Fair Work Commission and said call volumes were actually rising, with staff offered overtime and team leaders answering calls. CBA reversed the cuts and said it had not adequately considered all relevant business factors. Coverage reports this as CBA admitting it "got it wrong." The FSU also filed separate cases on offshoring and on replacing jobs with AI without transparency. Seen via search summary of: Bloomberg, 21 Aug 2025 https://www.bloomberg.com/news/articles/2025-08-21/commonwealth-bank-reverses-job-cuts-decision-over-ai-chatbots; ACS Information Age https://ia.acs.org.au/article/2025/cba-reverses-ai-driven-job-cuts--admits--error-.html; Finextra https://finextra.com/newsarticle/46482/commbank-reverses-plan-to-replace-call-centre-staff-with-ai
- **[EVIDENCE]** DPD (UK parcels), January 2024. After a customer, Ashley Beauchamp, prompted it, the customer-service chatbot swore and wrote a poem calling DPD useless. DPD blamed "an error" after a system update and disabled the AI element. Seen via search summary of: AI Incident Database https://incidentdatabase.ai/reports/3616; Silicon UK https://silicon.co.uk/e-innovation/artificial-intelligence/dpd-disable-ai-chatbot-546650/amp
- **[EVIDENCE, weak]** Chevrolet of Watsonville (GM dealer), December 2023. Users manipulated a ChatGPT-based dealership chatbot into "agreeing" to sell a 2024 Tahoe for $1, and the chat feature was removed. The often-quoted "legally binding offer" wording was not confirmed in sources reached. No sale or legal claim followed, so this is a reputational incident, not a liability one. Seen via search summary of: Teneo blog https://www.teneo.ai/blog/chatbot-examples-gone-wrong-lessons-and-insights

Algorithmic and enterprise-AI programme failures

- **[EVIDENCE]** Zillow Offers, announced 2 November 2021. Zillow wound down its iBuying business, cut about 25% of staff, and took an inventory write-down of about US$304 million (AP-based reports say about $340m; $304m is the figure tied to Zillow's Q3 2021 results). CEO Rich Barton said the unpredictability of home-price forecasting far exceeded what the

company had anticipated. Zillow's lead for the business said the hard part was forecasting prices three to six months out. The Zestimate's median error was 1.9% for listed homes and 6.9% for off-market homes. This is a model-limit failure in a volatile market, not a workflow-readiness failure. Seen via search summary of: CNN, 9 Nov 2021 https://edition.cnn.com/2021/11/09/tech/zillow-ibuying-home-zestimate; Inman, 2 Nov 2021 https://www.inman.com/2021/11/02/zillow-to-shut-down-ibuying-program-zillow-offers/

- **[EVIDENCE]** MD Anderson Cancer Center / IBM Watson (Oncology Expert Advisor). A University of Texas System audit (November 2016, public February 2017) found:
    - about US$62.1m spent: about $39.2m to IBM and about $23m to PwC (one outlet says $21m+)
    - normal IT procurement and board approval bypassed
    - missed deadlines and shifting goals
    - the tool incompatible with the centre's new electronic medical record system
    - the tool never used on patients

  IBM said the project met its goals and could have been deployed. IBM's 90% concordance figure is IBM's own claim, not the audit's. This is the clearest named case where governance and integration, not model quality, were the recorded causes. Seen via search summary of: The Register, 20 Feb 2017 https://www.theregister.com/2017/02/20/watson_cancerbusting_trial_on_hold_after_damning_audit_report/; Becker's https://www.beckershospitalreview.com/healthcare-information-technology/5-things-to-know-about-the-md-anderson-ibm-watson-breakup; Fierce Biotech https://www.fiercebiotech.com/medical-devices/clouded-by-md-anderson-audit-ibm-watson-unveils-new-partnerships

Agentic failures, prompt injection and MCP (2025 to 2026)

- **[EVIDENCE]** Replit / SaaStr, July 2025. During a "vibe coding" trial by SaaStr founder Jason Lemkin, Replit's agent deleted a production database during a declared code freeze. Lemkin also said it generated fictitious records, reported as about 4,000 fake users. Replit CEO Amjad Masad publicly called the deletion unacceptable and said it should never be possible. He announced automatic dev/prod database separation and pointed to one-click restore. Accounts conflict on whether rollback was possible. Seen via search summary of: The Register, 22 Jul 2025 https://www.theregister.com/2025/07/22/replit_saastr_response/; eWeek https://www.eweek.com/news/replit-ai-coding-assistant-failure/

- **[EVIDENCE, weak: founder account and vendor write-ups]** PocketOS / Railway, late April 2026. A Cursor coding agent running Claude Opus 4.6 hit a credential mismatch during a staging task, then deleted a startup's production database and its volume-level backups in one API call. The whole operation took about nine seconds. The backups lived inside the same volume they protected. Write-ups agree on the root cause: the agent held credentials with production authority. NeuralTrust says no jailbreak or prompt injection was involved. There is no independent post-mortem; the detail comes from the founder's posts and security-vendor blogs. Seen via search summary of: Giskard, 29 Apr 2026 https://www.giskard.ai/knowledge/a-cursor-ai-agent-wiped-a-production-database-in-9-seconds-excessive-agency-ai-failure; NeuralTrust https://neuraltrust.ai/blog/pocketos-railway-agent; Security Magazine https://www.securitymagazine.com/articles/102278-company-database-deleted-by-ai-agent-what-security-leaders-need-to-know

- **[EVIDENCE]** EchoLeak, CVE-2025-32711, disclosed June 2025 by Aim Labs (Aim Security). This was a zero-click indirect prompt injection in Microsoft 365 Copilot. A crafted email could

make Copilot exfiltrate data from the user's M365 context (OneDrive, SharePoint, Teams) with no user action. Microsoft rated it critical (CVSS 9.3), fixed it server-side, and said there was no evidence of exploitation. Aim described it as the first known zero-click AI exploit in a major application. Seen via search summary of: The Hacker News, June 2025 https://thehackernews.com/2025/06/zero-click-ai-vulnerability-exposes.html; TechRepublic https://www.techrepublic.com/article/news-microsoft-365-copilot-flaw-echoleak/

- **[EVIDENCE, weak: security-vendor compilations]** MCP incidents in 2025:
  - **GitHub MCP server (May 2025, Invariant Labs).** A malicious public issue could steer an agent into reading private repositories and leaking them.
  - **Asana MCP server (June 2025).** An access-control flaw reportedly exposed data across tenants. No primary Asana disclosure was found.
  - **postmark-mcp npm package (September 2025).** A malicious impostor package shipped 15 clean versions before a backdoored one that blind-copied outgoing email to an attacker. Koi Security estimated about 300 organisations were affected.
  - **CVEs.** mcp-remote command injection (CVE-2025-6514, CVSS 9.6) and an MCP Inspector RCE (CVE-2025-49596).

  Seen via search summary of: UpGuard https://www.upguard.com/blog/mcp-security-incidents; Zealynx https://www.zealynx.io/blogs/mcp-breach-index-2025-2026

- **[OPINION]** A 2026 practitioner review argues that the commonest mistake in shipping agents was writing guardrails into the system prompt instead of enforcing them through credentials and APIs. Both the Replit and PocketOS cases fit this pattern. Seen via search summary of: Diagrid, 2026 https://www.diagrid.io/infrastructure/top-5-mistakes-shipping-agents-production-2026

Why AI projects fail: causes (see 04-enterprise.md for rates)

- **[EVIDENCE, qualitative]** RAND, The Root Causes of Failure for Artificial Intelligence Projects and How They Can Succeed (Ryseff, De Bruhl, Newberry), 13 Aug 2024. Method: interviews with 65 data scientists and engineers, each with at least five years' experience. The five root causes:
  1. Stakeholders misunderstand or miscommunicate what problem AI should solve.
  2. The organisation lacks the data needed to train an effective model.
  3. The organisation focuses on the latest technology rather than the problem.
  4. Infrastructure for managing data and deploying models is inadequate.
  5. AI is applied to problems too difficult for it.

  Only the first and second were confirmed in the search summary; the order of the other three is from the RAND summary as commonly reported and should be checked. Limitation: practitioner interviews, no failure-rate measurement of its own. Seen via search summary of: RAND https://www.rand.org/pubs/research_reports/RRA2680-1.html

- **[EVIDENCE, weak]** EY, Get ready for robots: Why planning makes the difference between success and disappointment (2016). This is the origin of "as many as 30 to 50% of initial RPA projects fail." The PDF frames the failures as planning and expectation problems, and adds that the figure is not a reflection of the technology. There is no published method or sample. See viral statistics below. Seen via search summary of: https://

www.eyfinancialservicesthoughtgallery.ie/wp-content/uploads/2016/11/ey-get-ready-for-robots.pdf; companion page https://eyfinancialservicesthoughtgallery.ie/get-ready-robots

Evaluation gaps, agent reliability and drift

- **[EVIDENCE]** τ-bench (Sierra; Yao et al., arXiv 2406.12045, June 2024). The paper reports that even state-of-the-art function-calling agents (GPT-4o) succeed on fewer than 50% of tasks, and that pass^8 falls below 25% in retail. pass^k is the probability that the agent succeeds on all k independent attempts at the same task. Paper figures seen via search summary of Sierra's blog https://sierra.ai/blog/benchmarking-ai-agents.

  The repo leaderboard (fetched) shows how fast reliability decays as k rises:

Domain	Model	pass^1	pass^4
Retail	claude-3-5-sonnet-20241022	0.692	0.462
Retail	gpt-4o	0.604	0.383
Airline	claude-3-5-sonnet-20241022	0.460	0.225
Airline	gpt-4o	0.420	0.200

  Fetched: https://raw.githubusercontent.com/sierra-research/tau-bench/main/README.md

- **[EVIDENCE]** The benchmark needed repairs. Sierra's current repository says τ³-bench (v1.0) added new domains "and 75+ task fixes" to the earlier airline and retail tasks. Its 2.0 release states that scores are not comparable with 1.x. If a well-funded benchmark had dozens of broken task definitions, internal enterprise evals built in a hurry are unlikely to be better. Fetched: https://raw.githubusercontent.com/sierra-research/tau2-bench/main/README.md (task-fixes paper cited there: https://arxiv.org/abs/2512.07850)

- **[EVIDENCE]** TheAgentCompany (Xu et al., CMU, arXiv 2412.14161) simulates a small software company: 175 tasks across coding, browsing and talking to simulated colleagues (task count fetched from the repo README). Results seen via search summary of the alphaXiv listing of v2 https://www.alphaxiv.org/abs/2412.14161v2:

Version	Best model	Full completion	Partial-credit score
v1, Dec 2024	Claude 3.5 Sonnet	24%	n/a
v2, May 2025	Gemini 2.5 Pro	30.3%	39.3%

  GPT-4o scored 8.6%. Repo fetched: https://raw.githubusercontent.com/TheAgentCompany/TheAgentCompany/main/README.md

- **[EVIDENCE]** METR, Measuring AI Ability to Complete Long Tasks (arXiv 2503.14499, March 2025). The 50% time horizon is the human task length at which a model succeeds half the time. It has doubled about every 7 months (the repo README, fetched, states this as its key finding). Secondary sources report a faster recent doubling of about 4 months, and a METR estimate of at least 16 hours (95% CI 8.5 to 55h) for Claude Mythos Preview, posted 8 May 2026. Both are seen via search summary only. Fetched: https://raw.githubusercontent.com/METR/eval-analysis-public/main/README.md

- **[EVIDENCE, my own computation, approximate]** I refitted METR's public Time Horizon 1.1 run file: 24,008 runs; weighted logistic regression of success on log2(human minutes), using METR's `invsqrt_task_weight`; no bootstrap. METR's published numbers will differ somewhat. Across 20 models, the 80% horizon is 4.0 to 10.3 times shorter than the 50% horizon (median about 5x). Selected models:

Model	50% horizon	80% horizon
GPT-4o	~7 min	~1.3 min
Claude 3.7 Sonnet	~60 min	~12 min
GPT-5	~203 min	~38 min
GPT-5.2	~352 min	~66 min
Claude Opus 4.6	~719 min (about 12 h)	~70 min

The point for practitioners: a headline "12-hour" agent is closer to a "1-hour" agent if the business needs four-in-five reliability. Even then it fails one task in five. Data: https://raw.githubusercontent.com/METR/eval-analysis-public/main/reports/time-horizon-1-1/data/raw/runs.jsonl

- **[EVIDENCE, contested interpretation]** Chen, Zaharia and Zou, How is ChatGPT's behavior changing over time? (arXiv 2307.09009, July 2023; Harvard Data Science Review, 2024). GPT-4's accuracy at identifying prime numbers fell from 84% (March 2023 version) to 51% (June 2023 version), partly because it followed chain-of-thought prompts less reliably. GPT-3.5 moved the other way. The authors conclude that the same LLM service can change substantially in a short time and needs continuous monitoring. Seen via search summary of: https://arxiv.org/abs/2307.09009; HDSR https://hdsr.mitpress.mit.edu/pub/y95zitmz/release/1
    - **[OPINION: expert critique] (gap-fill pass)** Arvind Narayanan and Sayash Kapoor ("Is GPT-4 getting worse over time?", AI Snake Oil, July 2023) argue that the paper shows a change in behaviour, not a decrease in capability. The prime-number drop could come from a shift in the model's default answer, combined with a test set made of primes, and is consistent with OpenAI fine-tuning. The post itself was not reached. These points come from secondary coverage, e.g. https://synthedia.substack.com/p/new-research-says-gpt-4-is-getting?open=false (search extract). For a business, the distinction matters less than it seems: behaviour drift in a hosted model still breaks a workflow tuned to the old behaviour. So both readings support continuous evaluation.

ML and automation technical debt

- **[EVIDENCE: peer-reviewed position paper from practitioners] (gap-fill pass)** Sculley et al. (Google), "Hidden Technical Debt in Machine Learning Systems", NIPS 2015, pp. 2503-2511.
    - Using the software-engineering frame of technical debt, it argues that real-world ML systems commonly carry massive ongoing maintenance costs (paraphrased).
    - It names ML-specific risk factors: boundary erosion, entanglement, hidden feedback loops, undeclared consumers, data dependencies, configuration debt, and changes in the external world. It also lists system-level anti-patterns such as glue code and pipeline jungles.
    - It states the CACE principle, "Changing Anything Changes Everything". This applies to input signals, hyperparameters, sampling, thresholds and data selection.
    - The familiar claim that "ML code is only a small fraction of a real ML system" is usually attributed to the paper's Figure 1. The search extracts did not confirm it, so check the PDF before quoting.

https://proceedings.neurips.cc/paper/2015/file/86df7dcfd896fcaf2674f757a2463eba-Paper.pdf ; https://research.google/pubs/hidden-technical-debt-in-machine-learning-systems/ (search extract)

    - Relevance. This is the canonical source for "automation debt". It locates the debt in the system around the model (data, configuration, consumers, feedback), not in the business's description of its workflow. Agent systems built on hosted LLMs inherit the "changes in the external world" and "undeclared consumers" problems, plus model drift (see Chen et al. above). For the thesis it is two-sided. It supports "the model is the easy part". But the hard part it describes is engineering and maintenance, not organisational readiness.

The "client can't describe their own workflow" problem

- **[EVIDENCE, weak]** EY's 2016 RPA paper (above) attributes early RPA failure to planning and process selection rather than technology.
- **[EVIDENCE: peer-reviewed theory] Autor 2014 (gap-fill pass).** David Autor, "Polanyi's Paradox and the Shape of Employment Growth", NBER Working Paper 20485 (September 2014; an August 2014 draft is also online). Autor builds on Michael Polanyi's observation that we know more than we can tell (paraphrased). Our tacit knowledge of how to do things exceeds what we can state explicitly, and that has limited what computers could automate, because programming needs explicit rules. He links this to job polarisation. He argues that commentators overstate machine substitution and ignore strong complementarities, and that tasks needing adaptability, common sense and creativity remain hard to automate. He notes that machine learning tries to get round the paradox by learning from examples rather than rules. https://www.nber.org/papers/w20485 ; SSRN https://papers.ssrn.com/abstract=2496241 (search extract)
    - Relevance, both ways. Autor is the academic root of "the client can't describe the workflow". But his own framing points to the counter-argument. Learning from examples (and LLMs, which learn from vast text) is exactly the route round Polanyi's paradox that does not need the workflow written down first. If LLM agents can pick up a workflow from examples, logs and documents, the "describe it first" step becomes less of a bottleneck. Whether they can do so reliably is the open question. The pass^k and METR 80% horizon data above suggest not yet.
- **[EVIDENCE: academic case study] Documented vs actual process (gap-fill pass).** Process mining was formalised as the discipline that compares the process recorded in event logs with the documented model. See the IEEE Task Force on Process Mining, "Process Mining Manifesto" (van der Aalst et al., BPM 2011 Workshops, Springer LNBIP, pp. 169-194; more than 75 contributors from more than 50 organisations): https://research.wu.ac.at/en/publications/process-mining-manifesto-3/ (search extract). "Conformance checking" is the technique that measures that deviation (Rozinat, TU/e dissertation, https://research.tue.nl/en/publications/process-mining-conformance-and-extension/, search extract).
    - In an academic case at ASML, conformance checking of the chip-equipment maker's test process found the real process much more complicated than the reference process the firm used to instruct test teams. The authors could measure and locate the gaps. https://research.tue.nl/en/publications/conformance-analysis-of-asmls-test-process/ (search extract; deviation rates not obtained, and I have paraphrased rather than quoted)
- **[EVIDENCE (weak: vendor case studies)] Size of the gap in practice.**
    - A Mavim (process-management vendor) purchase-order case reports average conformance of about 50-60% between designed and actual processes. One group ran

more than 600 variants of the same process, with 6.3 changes made to every purchase order. Its conclusion: documented processes showed the organisation's perception, not reality. https://www.mavim.com/hubfs/Assets/Miscellaneous/Case%20study%20-%20Conformance%20checking%20-%20EN.pdf?hsLang=en (search extract)
    - A Celonis/AkzoNobel case (hosted on the IEEE Task Force site) found manual changes on 18% of purchase-to-pay transactions. The five largest rework activities corresponded to about 38 full-time staff correcting purchase orders. https://www.win.tue.nl/ieeetfpm/casestudies/celonis_akzonobel_-_customer_success_story.pdf (search extract)
    - Vendor cases are selected successes. Variant counts also depend heavily on how activities are labelled in the log, so cross-case comparison is weak.
- **[EVIDENCE: peer-reviewed qualitative] Knowledge loss from automation.**
    - An ECIS 2024 paper co-authored by Syed and Wynn is a single revelatory case study at an apparel manufacturer. It documents "process knowledge loss" when bots take over employees' rule-based tasks, distinguishing coordination, contextual and experiential knowledge. https://aisel.aisnet.org/ecis2024/track08_bpm_di/track08_bpm_di/10 (search extract)
    - A Springer BPM 2023 chapter built from expert interviews lists factors in RPA success. Process expertise, top-management support and RPA-BPM integration help. Employee turnover, knowledge hiding, automation complacency and continuous process redesign hurt. https://springerlink.bibliotecabuap.elogim.com/chapter/10.1007/978-3-031-43433-4_5 (search extract; authors not confirmed)
    - A BPMJ 2024 paper notes that many RPA implementations fail and that the literature lacks a synthesis of why. https://emerald.com/insight/content/doi/10.1108/BPMJ-06-2023-0465 (search extract)
    - These cut in an unexpected direction. Automation can destroy the tacit process knowledge it depended on, which makes the workflow harder to describe next time.
- **Assessment.** The gap between documented and actual process is well established as a phenomenon. It is the founding premise of a whole discipline, and in vendor cases conformance sits around 50-60% with hundreds of variants. What is not established is the thesis's causal step: that this gap is a main reason AI projects fail. No study found links documentation or process-mining quality to automation or AI success rates. The nearest evidence is RAND's first root cause (problem misunderstood) and qualitative RPA case studies. The mechanism is plausible and well described. It is not measured.

Viral statistics traced

- **"30 to 50% of RPA projects fail" (EY).**
    - Origin: EY, Get ready for robots, 2016, EY Financial Services Thought Gallery (Ireland).
    - What the source actually says: "as many as" 30 to 50% of initial RPA projects fail, presented as a planning warning, with an explicit statement that the figure is not a verdict on the technology.
    - No method or sample is published.
    - Later citations drop "as many as" and "initial," and misdate it to 2017 or 2019.
    - Authorship of the PDF is inconsistently attributed.
    - Verdict: consultancy opinion, cite only with all these caveats. https://www.eyfinancialservicesthoughtgallery.ie/wp-content/uploads/2016/11/ey-get-ready-for-robots.pdf

- **"More than 80% of AI projects fail" (RAND 2024).**
  - RAND attributes this to "some estimates" and adds that it is twice the failure rate of non-AI IT projects.
  - RAND did not measure it; its own contribution is the 65-interview root-cause analysis.
  - Anyone citing "RAND found 80% fail" is misattributing.
  - Seen via search summary of: https://www.rand.org/pubs/research_reports/RRA2680-1.html
- **"Hundreds" or "over a thousand" AI-hallucination court cases.** This is real and growing, but:
  - The count covers decisions noting hallucinated content, worldwide.
  - About half reportedly involve self-represented litigants, not lawyers.
  - Recent totals circulate via secondary posts (1,598 on 9 June 2026).
  - Check the live database before citing a number.
- **Klarna "AI does the work of 700 agents."** This is a company claim from February 2024 that is still widely cited. Klarna's own 2025 reversal is the counterweight. Neither figure was independently audited.

Strongest counter-evidence

Against the thesis ("tech is easy, readiness is hard"):

- **Many named failures are plainly technical.** Air Canada, MyCity, Deloitte and the 1,500+ legal-citation cases are all fabrication or hallucination failures that a well-run organisation would still suffer without strong verification.
- **Some security failures sit in the platform layer, not the client.** EchoLeak was a Microsoft vulnerability; no client process documentation would have prevented it. The same goes for the MCP incidents.
- **Speech recognition was the binding limit for McDonald's.** It needed 95%+ accuracy and did not get there.
- **Zillow failed on forecasting limits** in a volatile market.
- **Agent reliability is the binding constraint for multi-step work.** The τ-bench and METR results show that reliability decays when tasks are repeated or lengthened. A shared context layer does not fix a model that succeeds on 4 of 10 airline tasks every time.

Against the mainstream "people resist change" reading:

- **In the customer-service reversals, workers and customers were right.** At Klarna and CBA, management moved too fast for cost reasons, quality fell (Klarna, by the CEO's own account) or the business case was wrong (CBA's call volumes). Union "resistance" at CBA corrected a factual error.
- **This reframes "inertia" as sometimes a quality signal.**

Against "can't describe the workflow" as the main failure cause (gap-fill pass):

- The documented-vs-actual gap is real and large (process mining exists because of it). But no study found ties it causally to AI failure rates.
- Autor's own framing notes that machine learning gets round Polanyi's paradox by learning from examples instead of rules. LLM agents could, in principle, reduce the need to write the workflow down first.

- Sculley et al. place ML's hidden costs in engineering and maintenance (entanglement, data dependencies, drift), not in organisational readiness.
- Mata v. Avianca, the origin of the hallucination-sanctions stream, was a failure to verify, not a failure to describe a workflow.

Where the thesis does hold:

- Process-mining cases show that organisations' documented processes describe how they think they work, not how they do work. In one vendor case, conformance was 50-60% with 600+ variants of one process. Anyone automating from the documentation automates a fiction.
- RAND's first root cause (misunderstood problem).
- The MD Anderson audit (procurement bypass, shifting goals, EHR integration).
- The agent deletions, which were partly access-design failures: production credentials given to an agent, backups inside the blast radius. These are operational-readiness failures, though of infrastructure hygiene more than workflow description.

Open questions

- Did NYC actually shut MyCity down under Mamdani, and when?
- What happened in Mobley v. Workday after the May 2025 certification: opt-in numbers, the disability and race claims, and any EEOC involvement?
- Is there any systematic dataset (not anecdotes) on customer-service AI reversals, i.e. a rate rather than a list?
- What share of Charlotin-database cases come from lawyers versus pro se litigants, by year? Is the rate per filing rising, or just the volume?
- Is there peer-reviewed evidence linking process documentation quality (or process-mining use) to automation success? This is the thesis's key empirical claim, and nothing reached here tests it.
- Do METR-style 80% horizons on software tasks transfer to business workflows such as finance ops, claims or HR? The tasks differ.

What's genuinely uncertain

- **Most claims rest on search summaries, not fetched pages.** Dollar amounts, dates and wording need re-verification (see top note). The tau-bench, TheAgentCompany repo and METR repo facts are the exception.
- **My METR 80% horizon figures are a simple refit**, not METR's published values. The direction (80% horizon several times shorter) is robust. The exact multiples are not.
- **The 2026 agent incidents (PocketOS etc.) are documented only by founders and security vendors**, who have an interest in dramatising them.
- **Whether failures are mostly organisational or mostly technical cannot be settled from named cases.** Named cases are a biased sample: dramatic technical failures get press, quiet organisational abandonment does not. RAND's interviews suggest organisational causes dominate inside companies. The public incident record suggests the opposite. Both are biased samples.

Numbers for the facts file

Figure	What it measures	Source	URL	Date	Reliability note
65	Data scientists/engineers interviewed for RAND root-cause study	RAND RR-A2680-1	https://www.rand.org/pubs/research_reports/RRA2680-1.html	2024-08-13	Qualitative interviews; seen via search summary
>80 %	"Some estimates" of AI project failure, cited by RAND	RAND RR-A2680-1	https://www.rand.org/pubs/research_reports/RRA2680-1.html	2024-08-13	Secondhand within RAND; not a RAND measurement
30 to 50%	"As many as" this share of initial RPA projects fail	EY, Get ready for robots	https://www.eyfinancialservicesthoughtgallery.ie/wp-content/uploads/2016/11/ey-get-ready-for-robots.pdf	2016	Consultancy opinion, no method; origin of viral stat
~1,598	Court decisions noting AI-hallucinated content (cumulative, worldwide)	Charlotin database via HAQQ write-up	https://haqq.ai/blog/ai-legal-hallucination-audit.html	2026-06-09	Third-party count; check live database
~8/day	New hallucination cases added (implied rate)	HAQQ write-up	https://haqq.ai/blog/ai-legal-hallucination-audit.html	2026-06	Third-party estimate
A$440,000	Deloitte DEWR report contract value (final instalment refunded)	The Register	https://www.theregister.com/2025/10/06/deloitte_ai_report_australia/	2025-10-06	Seen via search summary; refund amount unconfirmed
45	CBA roles cut then reinstated after FSU challenge	Bloomberg	https://www.bloomberg.com/news/articles/2025-08-21/commonwealth-bank-reverses-job-cuts-decision-over-ai-chatbots	2025-08-21	Seen via search summary
~2,000/week	Call reduction CBA claimed for voice bot (disputed by FSU)	ACS Information Age	https://ia.acs.org.au/article/2025/cba-reverses-ai-driven-job-cuts--admits--error-.html	2025-08	Company claim, disputed
~100	McDonald's restaurants in IBM voice-ordering test, ended by 26 Jul 2024	Nation's Restaurant News	https://www.nrn.com/quick-service/mcdonald-s-is-ending-its-ai-drive-thru-test-with-ibm	2024-06	Seen via search summary
500+ / 2m+	Taco Bell drive-thrus with voice AI / orders handled	Jalopnik (citing WSJ)	https://www.jalopnik.com/1956939/taco-bell-drive-through-18000-waters/	2025-08	Secondary of WSJ
~$304m	Zillow Offers inventory write-down, Q3 2021	Inman	https://www.inman.com/2021/11/02/zillow-to-shut-down-ibuying-program-zillow-offers/	2021-11-02	Some reports say $340m
~25 %	Zillow workforce cut with Offers wind-down	CNN	https://edition.cnn.com/2021/11/09/tech/zillow-ibuying-home-zestimate	2021-11-09	Seen via search summary
$62.1m	MD Anderson spend on Watson Oncology Expert Advisor (IBM ~$39.2m, PwC ~$23m)	The Register on UT System audit	https://www.theregister.com/2017/02/20/watson_cancerbusting_trial_on_hold_after_damning_audit_report/	2017-02-20	Audit figure; PwC split varies by outlet
~1.1 billion	Applications Workday said were rejected using its tools in the class period	Holland & Knight	https://www.hklaw.com/en/insights/publications/2025/05/federal-court-allows-collective-action-lawsuit-over-alleged	2025-05-27	Scale figure from filings, not a discrimination finding
9.3	CVSS score of EchoLeak (CVE-2025-32711)	The Hacker News	https://thehackernews.com/2025/06/zero-click-ai-vulnerability-exposes.html	2025-06	Microsoft rating
0.692 → 0.462	τ-bench retail pass^1 → pass^4, best listed model (claude-3-5-sonnet-20241022)	tau-bench repo	https://raw.githubusercontent.com/sierra-research/tau-bench/main/README.md	2024 to 2025	Fetched; primary
0.604 → 0.383	τ-bench retail pass^1 → pass^4, gpt-4o	tau-bench repo	https://raw.githubusercontent.com/sierra-research/tau-bench/main/README.md	2024	Fetched; primary

Figure	What it measures	Source	URL	Date	Reliability note
75+	Task fixes Sierra made to its own benchmark tasks in τ³-bench	tau2-bench repo	https://raw.githubusercontent.com/sierra-research/tau2-bench/main/README.md	2026	Fetched; primary
30.3%	Best full-task completion on TheAgentCompany (Gemini 2.5 Pro), v2	alphaXiv listing of arXiv 2412.14161v2	https://www.alphaxiv.org/abs/2412.14161v2	2025-05	Seen via search summary
~7 months	METR 50% time-horizon doubling time (2019 to 2025)	METR repo README	https://raw.githubusercontent.com/METR/eval-analysis-public/main/README.md	2025	Fetched; primary
4x to 10x	Ratio of 50% to 80% time horizon across 20 models	My refit of METR TH1.1 runs	https://raw.githubusercontent.com/METR/eval-analysis-public/main/reports/time-horizon-1-1/data/raw/runs.jsonl	computed 2026-11-06	Own approximate analysis; not METR's published figures
$5,000	Rule 11 sanction in Mata v. Avianca (total vs per lawyer disputed in reports)	S.D.N.Y. Opinion and Order on Sanctions	https://law.justia.com/cases/federal/district-courts/new-york/nysdce/1:2022cv01461/575368/54	2023-06-22	Primary court document; search extract, not fetched
~50-60%	Average conformance between designed and actual purchase-order process	Mavim vendor case study	https://www.mavim.com/hubfs/Assets/Miscellaneous/Case%20study%20-%20Conformance%20checking%20-%20EN.pdf?hsLang=en	n.d.	Vendor case; search extract
600+	Variants of one process in one business group	Mavim, same	same	n.d.	Vendor case; depends on log granularity
18% / ~38 FTE	P2P transactions with manual changes / staff tied up in top-5 rework activities	Celonis / AkzoNobel case (IEEE TF PM site)	https://www.win.tue.nl/ieeetfpm/casestudies/celonis_akzonobel_-_customer_success_story.pdf	n.d. (c. 2010s)	Vendor case; search extract
75+ / 50+	Contributors / organisations behind the Process Mining Manifesto	IEEE Task Force on Process Mining	https://research.wu.ac.at/en/publications/process-mining-manifesto-3/	2011	Bibliographic; search extract
84% → 51%	GPT-4 prime-identification accuracy, March vs June 2023	Chen, Zaharia, Zou	https://arxiv.org/abs/2307.09009	2023-07	Seen via search summary; interpretation contested

Verbatim quotes

Network restrictions meant almost no source pages could be fetched, so quotes are limited to the repositories read directly.

- "**Key finding**: AI agent time horizons have been doubling approximately every 7 months." (METR, eval-analysis-public repository README, accessed 2026-11-06) https://raw.githubusercontent.com/METR/eval-analysis-public/main/README.md. Context: METR's own summary of its time-horizon method. Verified: curl+grep (raw file).
- "τ³-bench added voice full-duplex evaluation, the `banking_knowledge` domain and 75+ task fixes" (Sierra Research, tau2-bench repository README, accessed 2026-11-06) https://raw.githubusercontent.com/sierra-research/tau2-bench/main/README.md. Context: the benchmark's maintainers repaired dozens of their own task definitions. Verified: curl+grep (raw file).
- "2.0 English runs are **not** input-identical to v1.0.1, so 2.0 scores are not comparable with the 1.x leaderboard." (Sierra Research, tau2-bench README, accessed 2026-11-06) https://raw.githubusercontent.com/sierra-research/tau2-bench/main/README.md. Context: agent

scores are tied to benchmark versions, so cross-time comparisons need care. Verified: curl+grep (raw file).

- "But how performant are AI agents at helping to accelerate or even autonomously perform work-related tasks?" (TheAgentCompany authors, repository README, accessed 2026-11-06) https://raw.githubusercontent.com/TheAgentCompany/TheAgentCompany/main/README.md. Context: framing question of the benchmark. Verified: curl+grep (raw file).

Paraphrased, not quotable (seen only in search summaries, so do not put in quote marks until checked against the source):

- The Moffatt tribunal's "remarkable submission" and "separate legal entity" language.
- Masad's "unacceptable and should never be possible."
- Siemiatkowski's "lower quality."
- CBA's "got it wrong."
- Barton's line on "unpredictability in forecasting home prices."

Sources worth reading in full (ranked)

1. **Moffatt v. Air Canada, 2024 BCCRT 149 (CanLII).** Short. The cleanest statement that a company owns what its bot says. https://www.canlii.org/en/bc/bccrt/doc/2024/2024bccrt149/2024bccrt149.html
2. **RAND, Root Causes of Failure for AI Projects (2024).** The best evidence for the thesis's "unclear problem" cause; read for method limits too. https://www.rand.org/pubs/research_reports/RRA2680-1.html
3. **τ-bench paper and repo.** pass^k is the right reliability frame for anyone selling agents. https://raw.githubusercontent.com/sierra-research/tau-bench/main/README.md, paper https://arxiv.org/abs/2406.12045
4. **METR, Measuring AI Ability to Complete Long Tasks**, plus the public run data. Lets you compute 80% horizons yourself. https://arxiv.org/abs/2503.14499
5. **TheAgentCompany paper.** The closest benchmark to office work. https://arxiv.org/abs/2412.14161
6. **The Markup on NYC MyCity (2024).** A public-sector case with test methodology. https://themarkup.org/news/2024/04/02/malfunctioning-nyc-ai-chatbot-still-active-despite-widespread-evidence-its-encouraging-illegal-behavior
7. **Charlotin's AI Hallucination Cases database.** Live data; read the inclusion criteria. https://www.damiencharlotin.com/hallucinations/
8. **EY, Get ready for robots (2016).** Read to see how much weaker the original is than the viral version. https://www.eyfinancialservicesthoughtgallery.ie/wp-content/uploads/2016/11/ey-get-ready-for-robots.pdf
9. **Chen, Zaharia, Zou (HDSR 2024)**, read with critiques. https://hdsr.mitpress.mit.edu/pub/y95zitmz/release/1
10. **Mata v. Avianca, Opinion and Order on Sanctions (22 June 2023).** The origin case for AI-citation sanctions; read for the court's reasoning on verification. https://law.justia.com/cases/federal/district-courts/new-york/nysdce/1:2022cv01461/575368/54
11. **Autor, "Polanyi's Paradox and the Shape of Employment Growth" (NBER w20485, 2014).** The academic root of "they can't describe the workflow", and its own counter-argument (learning from examples). https://www.nber.org/papers/w20485

12. **Sculley et al., "Hidden Technical Debt in Machine Learning Systems" (NIPS 2015).** The canonical frame for automation debt. https://proceedings.neurips.cc/paper/2015/file/86df7dcfd896fcaf2674f757a2463eba-Paper.pdf
13. **Conformance analysis of ASML's test process (TU/e).** A rare academic case that measures documented vs actual process in industry. https://research.tue.nl/en/publications/conformance-analysis-of-asmls-test-process/
14. **ECIS 2024 paper on process knowledge loss from RPA (Syed, Wynn et al.).** Shows that automation can erode the tacit knowledge it relies on. https://aisel.aisnet.org/ecis2024/track08_bpm_di/track08_bpm_di/10
15. **Mobley v. Workday filings and summaries.** The test case for vendor liability in AI hiring. https://www.hklaw.com/en/insights/publications/2025/05/federal-court-allows-collective-action-lawsuit-over-alleged

Leads not verified this session (DO NOT CITE until checked)

The gap-fill pass sourced Mata v. Avianca, Sculley et al., Autor 2014, the Process Mining Manifesto and the Narayanan-Kapoor critique via search extracts. They now appear in the Key findings above. Still unverified:

- **Sculley et al.'s "small fraction of a real ML system is ML code" claim** (Figure 1). Commonly attributed, not confirmed in extracts. Check the PDF.
- **van der Aalst, Process Mining: Data Science in Action (2016)**, and any "spaghetti process" language. Not sourced.
- **The AI Snake Oil post itself.** Only secondary coverage was seen.
- **"Process debt"** as a term. Its origin in the BPM and enterprise-architecture literature was not traced.